VACTOR 2100 GLOBAL

Combination Sewer Cleaner



VACTOR®
Subsidiary of Federal Signal Corporation

The Vactor 2100 Global Series is not available for sale in the US and Canada

VACTOR®2100 GLOBAL®
TECHNICAL DESCRIPTION / WALK-AROUND

GENERAL OVERVIEW

Combination Sewer Cleaner

Vactor 2100 Series combination sewer cleaners combine high-pressure water jetting and a high-flow vacuum source to scour pipes clean then vacuum up the material causing blockages to restore and maintain normal sewer flow.

Vactor products feature the exclusive Vactor JetRodder® water pump with jack hammer action. The powerful vacuum source can be either a centrifugal compressor (fan) or positive displacement blower (PD) depending on application. With the Modul-Flex design, numerous-sized debris tanks from 3 to 15 cubic yards (2.3 – 11.5 cu m), and water capacities from 500 to 1,500 gallons (1,893 – 5,678 liters) are available.







DEBRIS BODY

Material

3/16" (4.7 mm) - single axle
1/4" (6.4 mm) - tandem axle
Ex-Ten Steel abrasion and corrosion resistant
Yield point: 50,000 psi (3,448 bar)
Tensile strength: 70,000 psi (4,827 bar)

Form

Cylindrical for maximum strength

Volumetric Capacities

5 yd3 (3.8 m3)
10 yd3 (7.6 m3)
12 yd3 (9.2 m3)
15 yd3 (11.5 m3)

Mounting

On an independent sub-frame separate from the chassis frame and mounted via a 3 point mounting system to allow flexing to occur without causing frame damage

Rear Door

Hinged at the top of the debris body
Flat style with four individual rod and receiver block type sequential door locking system
Door has hydraulic locks and door power up, power down, open/close feature
Equipped with a replaceable neoprene seal to prevent leakage

Body Dump

The body raises 50° utilizing a forward mounted, double acting hydraulic dump cylinder

Standard Dump Height**

Allows easy body dumping at transfer stations or on containers
Single rear axle configuration: 48" (1.2 m)
Dual rear axle configuration: 60" (1.5 m)



High Dump





**May vary slightly according to chassis selection

DEBRIS BODY (CONTINUED)

Body Dump, Rear Door and Locking System Controls

Provided at a central curbside location, directly behind the cab of the truck conveniently located inside a sealed, lockable tool box

Body Deflector Plates

Two plates assist in the settlement of the debris in the body

Load Level Indicator

Mechanical of three positions, shows when the body is loaded to capacity

Dual Air Conveyance System

Air is drawn from two separate ports in the debris body to achieve the best possible material separation and minimize material being discharged to the atmosphere

The body incorporates a dual air ducting system and a dual stainless steel ball float shutoff system with high impact corrosion resistant screens, which slide out and are easily accessible for maintenance and cleaning without the need of special tools

Centrifugal Separation System

Four centrifugal separators (cyclones) mounted in line between the debris body and the vacuum system aid in the process of material separation and help to prevent the ingestion of 50 micron or larger particulate into the vacuum generator (centrifugal compressor or blower)



WATER TANK

Material

High quality aluminum

Does not require internal coatings

Easy repairable and patched if required

Form

Side tanks: cylindrical

Standard Capacities**

1000 gal US (3,785 l)

1300 gal US (4,921 l)

1500 gal US (5,678 l)

Location

Lengthwise, at or below the chassis frame rails for operational safety and to optimize weight distribution

Mounting

On the module sub-frame

They remain fixed and do not rise with the debris tank upon dumping, which allows for greater stability of the unit and avoids unnecessary wear and tear of the hydraulic system

The side tanks are easily removed for repair and maintenance, if required

Independence

The water tanks are totally separate from the debris tank

They do not provide any structural support to the debris tank nor share with it any common walls to prevent corrosion



** Consult with you authorized dealer about available capacities



WATER TANK (CONTINUED)

Filling System

The water tanks are adequately vented and have 6" (152 mm) connections between them for rapid and complete filling

Includes an anti-siphon device

Includes 25' (7.6 m) of hydrant fill hose and fittings

Includes two water level sight gauges, one located on each side of the unit

Includes gravity fill system

Includes a 2""Y" pattern filter with an 80 mesh stainless steel strainer to filter water before it enters into the water tank

Baffles

Across the water tanks to provide superior stability while in transit

VACUUM SYSTEM

CENTRIFUGAL COMPRESSOR

Vacuum Generation

Single stage: by one 38" (965 mm) dia. tapered centrifugal compressor

Double stage: by two 38" (965 mm) dia. tapered centrifugal compressors

Material

Aluminum for operator safety

Design

Riveted type

Thickness of Radial Blades

¼" (6.35 mm)

Material of Outer Housing

¼" (6.35 mm) spun steel

Outer Housing Drain

Includes a 2" (50 mm) drain, plumbed away from the fan for safe and easy cleanout

Centrifugal Compressor Drive

Direct, through a helical gear type step-up transmission drive with a step-up ratio of 1.6 to 1

Gear Drive

The gear drive attaches directly to the rotor shaft without the use of multiple stage V-belts or jack shafts

Lubrication

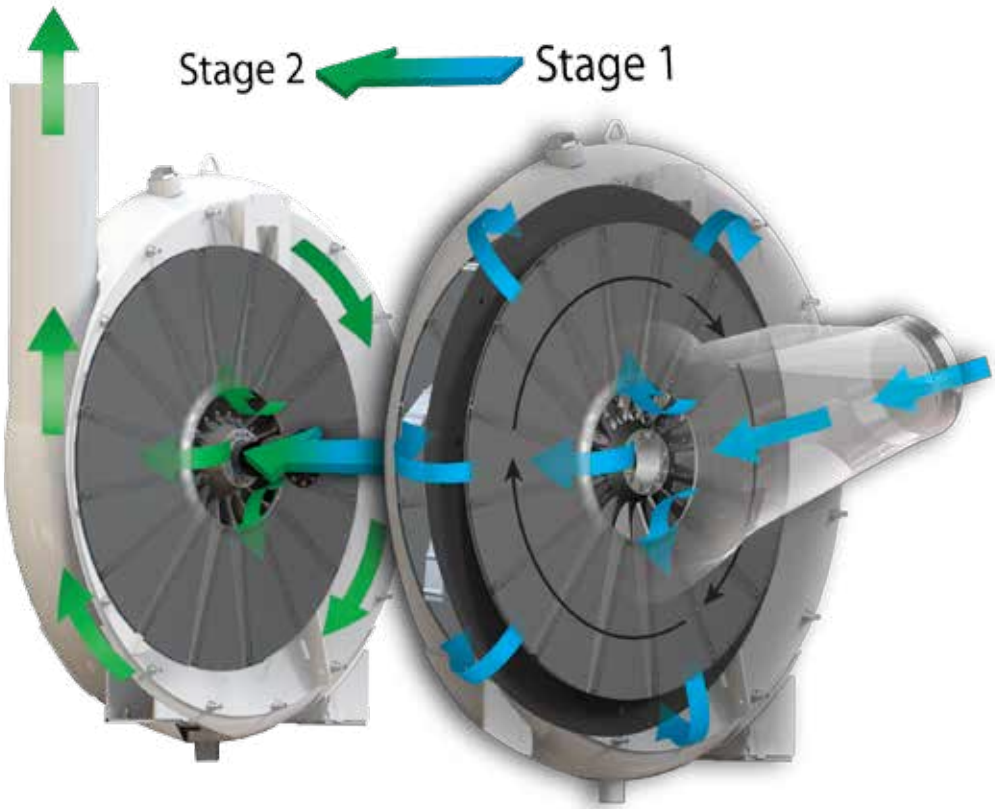
The gears and bearings are lubricated with a splash lubrication system, requiring no manual greasing

Drive Shaft Support

Via tapered roller bearings and gears

Fluid Coupler

Includes a fluid coupler that permits the engagement of the fan without the use of a mechanical clutch mechanism





Front of Fan



Back of Fan

VACUUM SYSTEM (CONTINUED)

CENTRIFUGAL COMPRESSOR AUXILIARY ENGINE

Auxiliary Engine

Industrial four cycle heavy duty engine
Liquid cooled, diesel powered, turbocharged

Cylinders

Four cylinders: single stage configuration
Six cylinders: double stage configuration

Horse Power

125 HP single stage, 185 HP dual stage

Displacement

276 in^3 (4.5 l) single

Shutdown System

- High temperature and low oil pressure
- Includes exhaust silencer

Support

On steel back rubber isolating mounts

Mounting

On its own frame mounted on the module sub-frame

Engine Shroud

Includes a noise deadening engine shroud mounted on roller tracts allowing for the shroud to easily be pulled away for compete engine access

Control Location

Conveniently located directly behind the cab

Controls

Tachometer with hour meter
Water temperature gauge
Oil pressure gauge
Voltmeter



VACUUM SYSTEM (CONTINUED)

POSITIVE DISPLACEMENT BLOWER

Vacuum Generation

By a positive displacement rotary lobe blower

Standard Capacities **

15"Hg

3500 CFM inlet volume at 0" Hg

3260 CFM at 15" Hg and 2250 rpm

16"Hg

4500 CFM inlet volume at 0" Hg

3620 CFM at 16" Hg and 2250 rpm

18"Hg

4500 CFM inlet volume at 0" Hg

3500 CFM at 18" Hg and 2250 rpm

Added Protection

Includes three vacuum reliefs set at rated vacuum capacity

Includes a stainless steel final filter screen and housing, located prior to the blower inlet

Blower Drive

From the chassis engine via the transmission drive shafts and a heavy-duty split shaft transfer case, direct to blower with no V-belts to maintain or adjust

Silencer

Horizontal, with exhaust above the cab, with rain cap

Air Shift Control

The blower, transfer case and hydraulic power are controlled by air shift control, located in the cab



** Consult your local dealer about other available capacities

VACUUM HOSE

Design

For front end operation with hose mounted and stored at the front mounted workstation

Eases positioning vacuum hose and allows for superior operational safety (minimizes the need to swing hose into traffic)

All connections between debris body and the vacuum system are of the self-adjusting pressure fitting type

Diameter of Hoses and Tubes

8" (203 mm) I.D.



BOOM

Location

Front mounted for superior safety and operational efficiency

Cab Protection

Includes a cab protection device with boom cradle, independent of cab, that protects the cab and prevents the boom to sway while in transit

Upon dumping, the boom and vacuum hose remain stationary and do not raise with the debris tank

Minimizes the possibility of running into power lines and low hanging branches

Material of Upper Debris Tube

Anchored steel for superior life

Rotation and Reach

The vacuum hose is mounted on a boom which rotates 180° and provides (8.4m) of reach off the centerline of the unit

Elbow

Made of steel with a 70° curve for superior air conveyance and to minimize excessive wear by friction and abrasion

Lift and Swing Movements

Up, down, left and right movements accomplished by cylinder actuated means

Activation

By an electric-over-hydraulic system which draws its power from the hydraulic pump driven from the chassis

Controls

At the front control station

At the boom pendant control that includes a safety emergency shutdown button

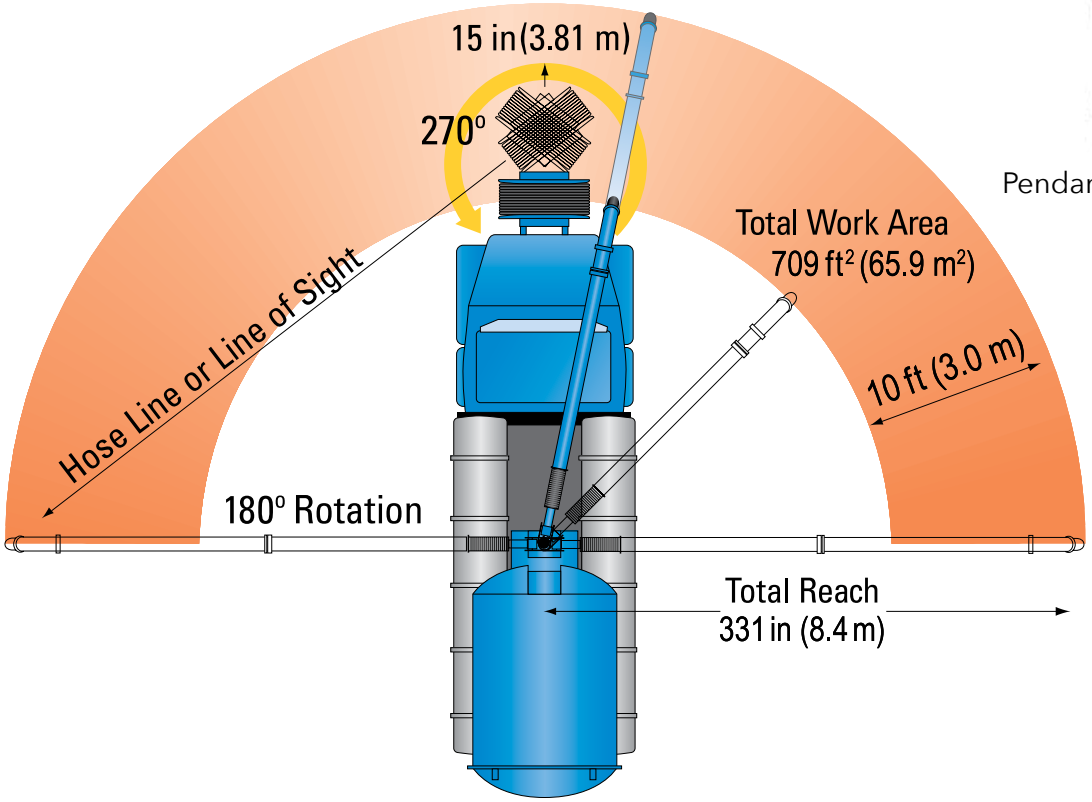
Standard Extension Pipes

The unit includes one set of aluminum pipe extensions that allows the operator to clean to depths up to 20' (6 m)

This set includes one 6'6" (1.98 m) catch basin nozzle with a steel reinforced tip and adjustable over-center quick clamps to join the supplied suction pipes



Pendant Boom Control



HIGH PRESSURE WATER PUMP

Vactor JetRodder® Pump

A positive displacement, high-pressure, double acting, hydraulically operated single piston pump

It has been specifically designed for sewer cleaning operations

Its design allows for configurations up to 100 gpm US (378 lpm) @ 3000 psi (206 bar)

Activation

Direct from the hydraulic pump driven from the chassis

Does not require an auxiliary engine or a belt drive

Oil to Water Ratio

One to one 1:1

Typical configurations in continuous duty operations

60 gpm US (227 lpm) @ 2000 psi (137 bar)
60 gpm US (227 lpm) @ 2500 psi (172 bar)
80 gpm US (302 lpm) @ 2000 psi (137 bar)
80 gpm US (302 lpm) @ 2500 psi (172 bar)

Cycles per Minute

60 gpm US (227 lpm) configuration: 14 cycles per minute

80 gpm US (302 lpm) configuration: 23 cycles per minute

"Jack Hammer" Action

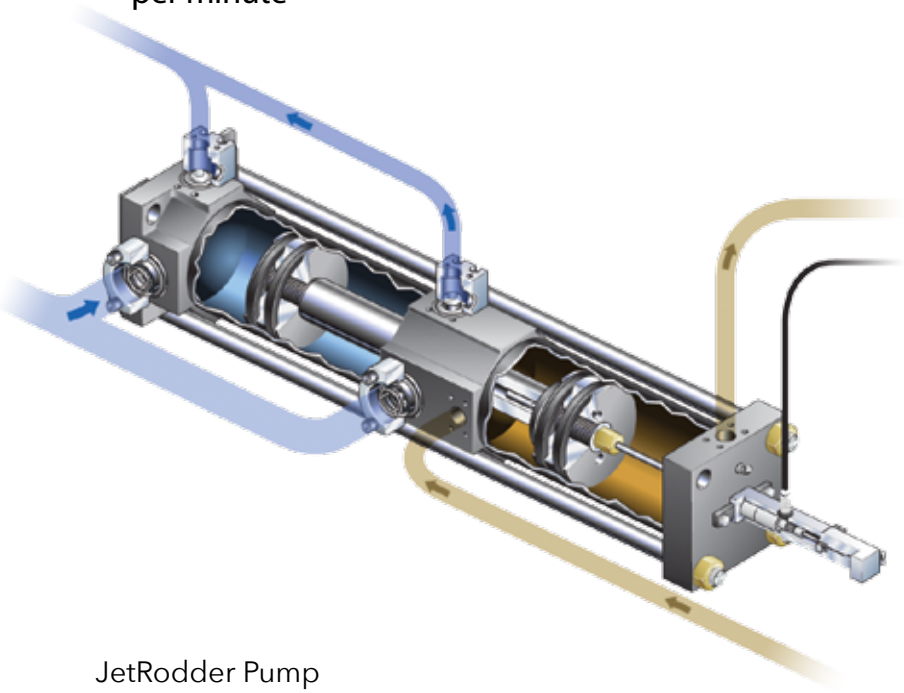
The pump's cycles create an automatic "Jack Hammer" action in the hose to assist the nozzle in breaking and in penetrating obstructions

Flooded Suction Inlet

The pump location provides a flooded suction inlet to eliminate potential cavitation damage



JetRodder Pump Location



JetRodder Pump



Water Controls

HIGH PRESSURE WATER PUMP (CONTINUED)

Operations Without Water

The pump has been designed so no damage will occur if it runs at normal operating pressure without water during 30 minutes

Vactor certifies this feature

Filter

Includes a 3" (76 mm) Y-type strainer with a 80 mesh stainless steel filter media and a 3" (76 mm) water shutoff valve located in the water pump suction line

Residual Water Purge

The pump allows for residual water to be purged out of the water system by pumping air through the entire system

Includes drain plugs at the bottom section of the water pump for complete draining

Simultaneous Operations

The water pump works independently from the vacuum system and can operate at maximum water pressure and flow while achieving maximum vacuum capability

Oil to Water Heat Exchanger

Included in the water system to cool all hydraulic fluids on the unit

Controls

For added control and safety the water pump can be engaged and disengaged from the driving mechanism at the front control panel by turning on or off an electric switch

Multi-flow System

The water system has the ability to vary flow from zero to its maximum configuration

Flow is varied by a hydraulically controlled system



Multiflow System

FRONT MOUNTED HOSE REEL

Mounting

The hose reel assembly is mounted on an independent frame that can be removed from brackets attached permanently to main truck frame members

Front mount is preferred for safety purposes as it allows the operator to avoid working before incoming traffic

Material

The hose reel is manufactured out of ¼" (6.35 mm) spun steel for added structural strength

Does not require internal or external reinforcements which could damage the rodder hose

Frontal Hydraulic Extension and Rotation

The hose reel is constructed to hydraulically extend forward 15" (381 mm) to provide full access to the chassis engine. The hose reel can telescope up to 15 inches (381 mm) forward and rotate up to 270 degrees to allow for optimal hose reel positioning without moving the truck.

The extension function is provided via a hydraulic cylinder, not a manual jack

Capacity

800' (25.4 mm) of 1" I.D. rodder hose



Front Mounted Hose Reel

FRONT MOUNTED HOSE REEL (CONTINUED)

Activation

Via a hydraulic motor with a dual sprocket and chain system that controls the reel speed and the forward and reverse rotations

Rotating Swivel Joint

Includes a 1" (25.4 mm) adjustable rotating swivel joint that has replaceable seals on the inlet line to provide free rotation of hose reel

Rotation Radius

The hose reel rotates 270º on its own center axle

The rotation angle is lower if the automatic hose guide option is added

Controls

All operating controls are located on the hose reel

Vactor certifies this feature



Rotating Swivel Hose Reel

HIGH PRESSURE HOSE

Diameter of Hoses and Tubes

1″ (25.4 mm) I.D.

Operating Pressure

2500 PSI

Burst Pressure

6250 PSI

Standard Length

400′ (121.9 m)

Nozzles

With tungsten carbide orifices for added life

The standard unit includes two jet nozzles:

- One 15° sand nozzle
- One 30° sanitary nozzle

Flexible Hose Guide with Restraining Rope

Included for protection of rodder hose and to provide help in locating nozzles in sewer lines

HYDRAULIC TANK

Design

The tank is a totally modular component and can be easily removed from its storage area located at the module's sub-frame

Capacity

67 gal US (253 l)

The hydraulic reservoir is baffled with a suction line filter

Includes inspection hatch for maintenance

Hydraulic Filter

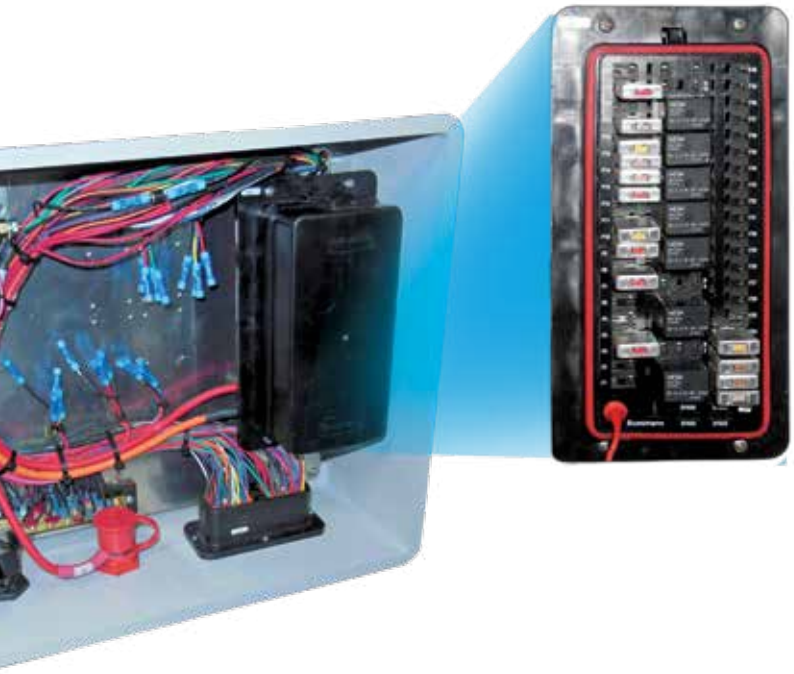
10-micron spin on type



High Pressure Handgun



Cleaning System with Handgun



CLEANING SYSTEM WITH HANDGUN

Description and Capacity

The system uses the JetRodder® water pump to supply up to 20 gpm US (75.7 l/min) at 600 PSI (41 bar) to the wash down handgun system, relieved to protect the operator

Handgun

Includes quick connect coupler

Allows changing the flow pattern from a fine mist to a steady stream

Hose

Includes a 35' (10.7 m) of ½" (152 mm) wire reinforced hose

Standard Connection

Includes a midship connection

Allows the operator to deliver water to area served by pick-up hose and to the inside of the debris body for clean out

Optional Connections

Front and rear

ELECTRICAL SYSTEM

Sealed System

The entire electrical system is vapor sealed to eliminate moisture damage to NEMA 4 (National Electrical Mfg. Assoc.) standards

Light Bulbs

Stop, tail, turn, clearance lights are LEDs

Wiring

Color coded and run in sealed terminal boxes

The electrical connections do not require exposed wires or terminals

FRONT OPERATION AND CONTROL STATION

Location

At the front of the truck mounted to the hose reel

Included Controls:

High pressure water pump on-off
Hose reel forward-reverse valve
Adjustable hose reel speed control
Oil dampened water pressure gauge
Hose reel extend/retract control
Blower tachometer with hour meter **
Vacuum gauge
Electronic E-Stop at each control station location
Boom controls

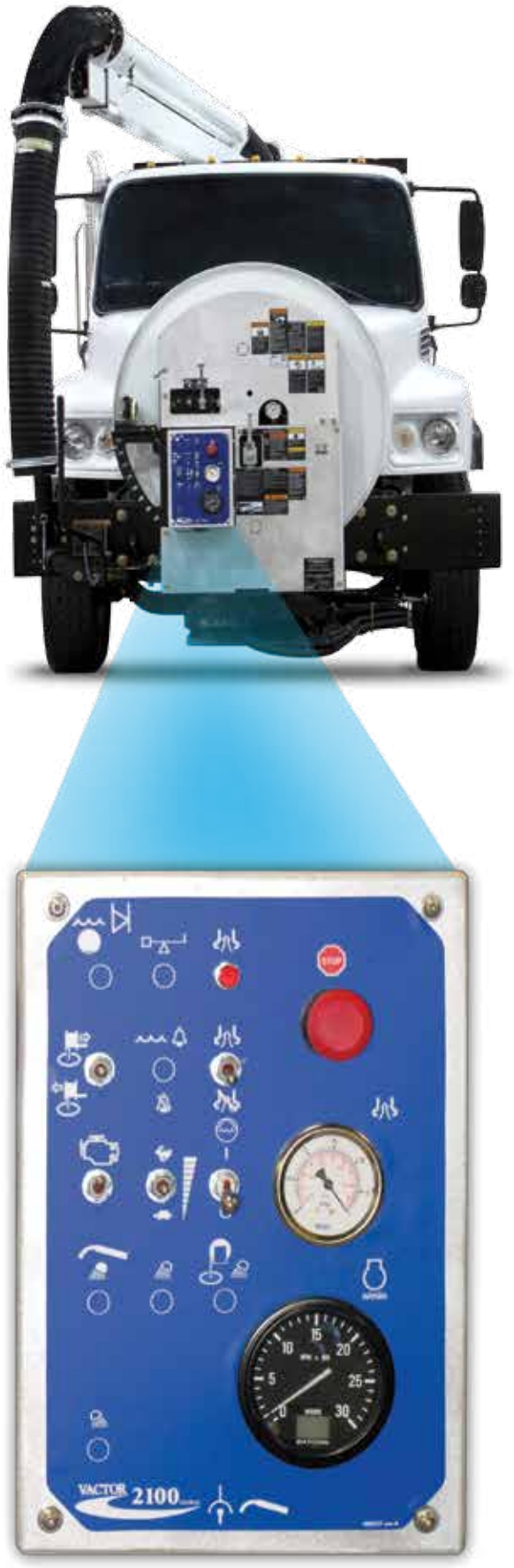
FINISH

Metallic Surfaces

Prior to Painting

- All metal surfaces are shot or sand blasted
- All surfaces are cleaned by phosphorous wash process
- Module primer
  2½ mm coat
- Module finish paint
  2½ mm coat

Module and components are painted prior to assembly process to avoid paint on hoses, fittings, electrical wires and connection



Operator Controls

** If applicable





For more than 100 years, Vactor Manufacturing has provided customers around the world with a wide range of material handling equipment.

Today, Vactor is the industry leader in vacuum excavators, sewer and catch basin cleaners and industrial vacuum loaders featuring innovative technology and custom configurable designs. With one of the largest and most developed distributor networks in the industry, Vactor has dealers located around the world.

Vactor Manufacturing, Inc. • 1621 South Illinois Street • Streator, IL 61364 U.S.A.
(815) 672-3171 Phone • (815) 672-2779 Fax

VACTOR
WORK ASSURED

Specifications subject to change without notice. Some products shown with optional equipment. Vactor® and JetRodder® are registered trademarks of Vactor Manufacturing. Vactor is a subsidiary of Federal Signal. Federal Signal is listed in the NYSE by the symbol FSS.

©2015 Vactor Manufacturing

Effective Date 3/15